

ADVANCED REVIEW

Weighted Ensemble Simulation: Advances in Methods, Software, and Applications

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ABSTRACT

For more than two decades, weighted ensemble (WE) path sampling strategies have enabled the simulation of pathways for rare events—or barrier-crossing processes—with significantly less computing cost than conventional simulations, all while preserving rigorous kinetics. This review highlights recent advances in WE methods and software, including tools for mechanistic analysis of path ensembles and efficient estimation of rates. We showcase successful WE applications across a wide range of condensed-phase processes, such as hybrid quantum mechanics/molecular mechanics (QM/MM) simulations of microsecond-timescale chemical reactions, and atomistic simulations of slower processes on the millisecond to seconds timescale. These applications span drug membrane permeation, ligand unbinding, and the large-scale opening of the SARS-CoV-2 spike protein. We also discuss the current limitations and key challenges facing WE strategies, which have yet to reach their full potential.

This article is categorized under:

Molecular and Statistical Mechanics > Molecular Dynamics and Monte-Carlo Methods

Software > Simulation Methods

Structure and Mechanism > Computational Biochemistry and Biophysics

1 | Introduction

The “weighted ensemble” (WE) path-sampling approach [1, 2] can be a powerful and unbiased tool for estimating rate constants and rare-event probabilities as well as quantifying mechanisms of transition processes. The strength of

WE derives from its use of multiple independent simulations, which are assigned weights and intermittently evaluated for continuation, replication, or pruning. Generically, WE uses replication (splitting) of trajectories to encourage sampling in rarer regions, while trajectory pruning (merging) reduces the use of computing resources in well-sampled areas of state

space. Although most often applied to molecular simulations, the WE strategy is very general and has also been applied to systems biology [3–5], weather modeling [6, 7], and astronomy [8].

The broad applications of WE in the literature reflect a key intrinsic feature of WE, namely, that trajectories are run fully independently for brief time intervals, and they are examined only at the end of these intervals. Because the trajectories are independent, excellent parallel scaling can be achieved, including on supercomputers [9, 10]. Arguably of equal importance, the intermittent examination of trajectories greatly facilitates interoperability. No modification of existing dynamics engines is necessary, in contrast to other path sampling methods [11–13]. This greatly facilitates changing the molecular dynamics engine or version, as well as allowing facile implementation of non-molecular simulations.

This review will focus primarily on the use of WE for molecular simulation. We tout the successes of WE but also are forthcoming about the method's limitations. Some limitations are intrinsic to complex systems, others to path sampling methods in general [14], and some are specific to WE. The challenges in applying WE to complex systems have motivated a large body of methodological improvements; fundamentally, WE is a “living” method, still under active development. The flexibility of WE allows for a wide range of methodological improvements and variations, which this review organizes for readers.

Our own experience derives almost exclusively from the WESTPA software, which is open source, highly scalable, portable, and interoperable with any stochastic dynamics engine

[9, 15]. However, we note that other WE implementations have also been reported [16, 17].

2 | The WE Approach

2.1 | The Basic Method

We now describe the traditional WE algorithm using static bins and continuous trajectories (Figure 1). Several variants that improve the power of WE will be reviewed in subsequent sections.

The WE method orchestrates a set of trajectories using groupings that typically are “bins” dividing up configuration space. Each trajectory i is assigned a weight w_i , and the weights remain normalized throughout a WE run, $\sum_i w_i = 1$, even though the number of trajectories may change due to split or merge events described below. Individual trajectory weights also change at split and merge events.

A WE simulation may be initiated from a single bin or a distribution of starting configurations in different bins, with any number of trajectory start points employed. Each bin is allocated a target number of trajectories with the qualitative goal of achieving an even distribution of sampling across configuration space.

In each iteration of a WE simulation, trajectories are run for a brief interval of time τ and then “resampled,” that is, split or merged (replicated or pruned) based on the bins (Figure 1). In a splitting event, typically triggered when a trajectory enters an

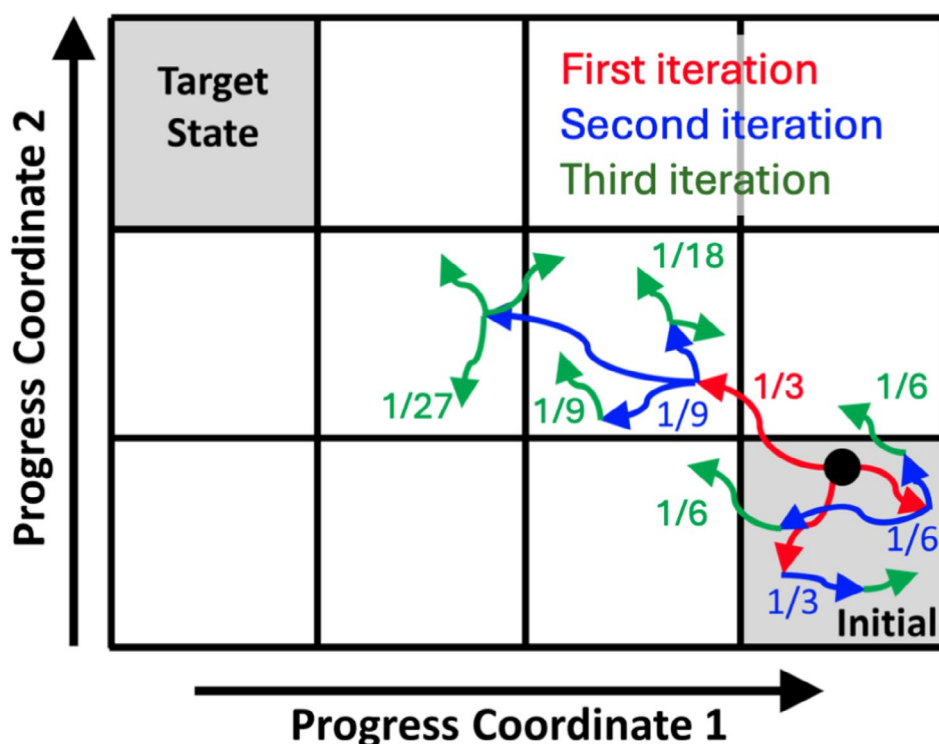


FIGURE 1 | The weighted ensemble (WE) strategy. This schematic illustrates a basic implementation of WE with fixed bins and a target of three trajectories per bin. Child trajectories equally share the weight from their parent, so weights sum to one in every iteration. Note that not every weight is indicated and that no merge events are depicted.

unoccupied bin, a “parent” trajectory is replicated into two or more “child” trajectories, with each child inheriting coordinates identical to the parent and an equal share of the parent’s weight to preserve normalization. Children also inherit the historical trajectory of their parents. By adding trajectories to newly visited regions of configuration space, WE enhances sampling of those areas (Figure 1).

Merging is the second type of resampling process. In a bin-based merging event, typically triggered when the number of trajectories exceeds the bin’s target number, one of a pair of trajectories i and j is chosen to proceed based on their weights. For example, trajectory i is selected with probability $w_i / (w_i + w_j)$ and assigned the new weight $w_i + w_j$ to maintain normalization, while trajectory j is not continued. The merge process can be repeated until target numbers are met.

After resampling by split/merge processes, the full WE step is repeated: dynamics of all trajectories for a time interval τ , followed by resampling (Figure 1). A WE run will continue for as many iterations as requested, hopefully enabling estimations of observables of interest.

2.2 | Notable Features of WE

2.2.1 | Simplicity and Interoperability

The ability of WE software, such as WESTPA [9, 15], to function easily with different dynamics engines, including tools outside the realm of molecular simulation, stems from the simplicity of the WE algorithm. WE only requires the ability to start, stop, and restart trajectories. Forces along progress coordinates do not need to be computed, as only unbiased dynamics are run. No “under the hood” modification of dynamics engines is necessary—ordinary command-line simulations are used. WE can be used with any stochastic dynamics engine and is equally valid for different thermodynamic ensembles such as “NVT” or “NPT,” that is, constant volume or pressure with fixed numbers of particles and temperature. The requirements are an ability to continue trajectories after checkpoints or pauses in an unbiased fashion and for trajectories to diverge from a common starting point due to a stochastic element of the dynamics (e.g., thermostat).

2.2.2 | Algorithmic Flexibility

There is considerably more flexibility in the general WE method than was just described for the traditional algorithm. Bin definitions can be changed at every iteration, as can bin allocations (target number of trajectories per bin); likewise, the value of τ can change at each iteration so long as it is the same for all trajectories [18]. Even the use of bins is not necessary, as resampling actually can employ any grouping of trajectories [19] because resampling based on WE weights does not modify the underlying distribution of trajectories on average. Beyond these variants on the basic WE elements, the sections below illustrate the almost unlimited flexibility in devising coordinates for WE bins or resampling, as well as the possibility to alter trajectory weights outside resampling for improved convergence.

2.2.3 | Rare Event Sampling via Continuous, Unbiased Trajectories

Unlike common methodologies that alter the free energy landscape to encourage the sampling of rare regions of phase space [20–23], WE uses standard simulation without artificial forces. As already described, WE only employs trajectory replication in important regions of phase space to encourage—but not force—transitions or other behaviors of interest. Because trajectory segments are continued from previous segments, the end result of a standard WE simulation is an ensemble of continuous trajectories from initial time to final; in the case of molecular simulation, the continuous trajectories can be analyzed like an MD simulation for any structural properties of interest.

2.2.4 | Statistical Exactness for Nonequilibrium and Equilibrium Properties

The WE method is unbiased in the sense that, with enough trajectories or when averaged over a large number of runs, the WE method will reproduce the same observables as would be obtained from launching a large number of ordinary trajectories from the same initial conditions [18, 24]. This holds for any property that can be computed from a path ensemble, such as nonequilibrium correlation functions, rate constants, or equilibrium observables obtainable from a fully relaxed ensemble. On the other hand, as with any simulation method, results obtained from finite WE simulations are subject to statistical noise. Substantial efforts, summarized below, have been devoted to estimating uncertainty in WE runs and overcoming sampling challenges specific to WE.

2.2.5 | The Use of Source–Sink Boundary Conditions for Rate-Constant Estimation

Kinetics computations are among those of highest interest for WE users, and WE enables rate-constant estimation via source-sink (aka “recycling”) boundary conditions [25]. The user defines the initial (source) and target (sink) states of interest, A and B: when trajectories reach the target B, they are recycled to the initial state A. In such a setup, the WE simulation will eventually reach a nonequilibrium steady state (NESS), enabling estimation of the rate constant k_{AB} from the probability “flux” arriving per unit time:

$$k_{AB} = \text{Flux}(A \rightarrow B | \text{NESS}) \quad (1)$$

a relation that does not hold in the transient relaxation regime prior to reaching NESS. See Figure 2. When the WE simulation does not fully relax to steady state, alternative analyses described below can be used for rate-constant estimation.

2.3 | Intrinsic Limitations of WE

The primary limitation of the WE approach stems from system’s intrinsic physical timescales and therefore is a challenge for any method that simulates a true path ensemble of continuous

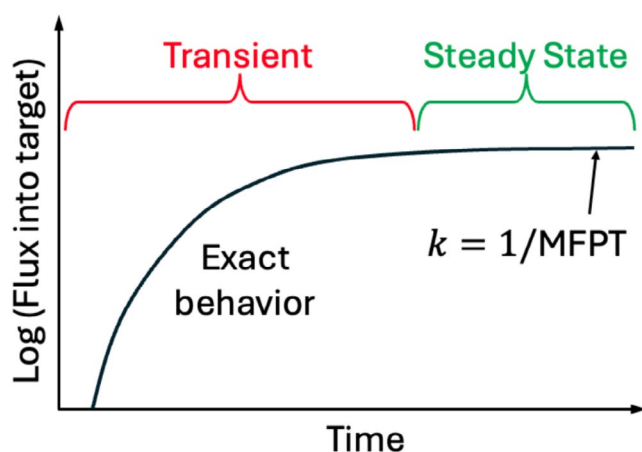


FIGURE 2 | Estimating the rate constant from WE flux into target state. For a typical WE run initialized with all probability outside the target state, the probability of arriving at the target per unit time (the “flux”) will increase during a transient period and eventually reach a steady state. Note that the transient period may be long compared to available resources, a challenge discussed below.

trajectories. Specifically, any transition process of interest can be characterized by an average event duration (aka “transition path time”), $\langle t_{TP} \rangle$ [26, 27]. Thus, an ensemble of $n \gg 1$ trajectories consists of a total time of $n \cdot \langle t_{TP} \rangle$, which represents the absolute minimum computational cost if correctly distributed transition trajectories could be generated fully independently (which is not possible). Therefore, an additional inefficiency factor $m > 1$ (likely $m \gg 1$) representing the overhead cost of generating independent trajectories also accrues, and the total cost for an ensemble is $m \cdot n \cdot \langle t_{TP} \rangle$, which does not account for potentially non-Gaussian large fluctuations in t_{TP} values [28]. Even a transition process for which $\langle t_{TP} \rangle \sim 10$ ns may require many μ s of trajectory data during WE or another method generating continuous path ensembles. Estimates for $\langle t_{TP} \rangle$ vary: $\sim 1 - 100$ ns for small-protein folding on the μ s to ms timescale [28, 29], ~ 5 ns for diffusion-controlled protein–protein binding on the μ s timescale [30], and ~ 100 ns for protein–ligand unbinding on the seconds timescale [31].

The inefficiency factor m stems from correlations among trajectories generated by a particular simulation method. In WE, that correlation arises from the fundamental splitting operation [2]. Child trajectories of the same parent share an identical history up to the point of splitting, making them highly correlated. The practical consequence of highly correlated trajectories can be high run-to-run variance in (unbiased) WE estimates of observables such as the rate constant (Figure 2). This correlation-based variance and inefficiency can be ameliorated to some degree, and is addressed below in further detail. We also note that correlations make uncertainty quantification (UQ) more challenging, as addressed below.

Overall, while WE is a powerful and rigorous method, it is not guaranteed to work well on every system. For example, the unbinding of a highly charged ligand from a protein receptor represents a particularly challenging stress test [32]; in contrast, a more tractable application involves the unbinding of uncharged ligands (Section 5.2). The intrinsic costs based on physical properties of the system are significant, not only for

WE, but also for any method that provides true ensembles of transition paths, and this can be true for coarse-grained models [33–35]. Correlation-based inefficiency is also intrinsic to path sampling.

The major practical challenge in setting up WE simulations is the selection of progress coordinates and binning schemes. Better binning schemes can accelerate sampling of rare events and reduce run-to-run variance in estimating observables, as discussed below. On the other hand, a poor choice of bins could cause WE to fail to generate a transition of interest. In a large, complex system, useful progress coordinates may be particularly difficult to design, and there is no universal recipe that can be followed. These points motivate a more extensive discussion of progress coordinates and binning in the next section.

3 | Advances in WE Methodology

Significant efforts to advance WE methodology have been undertaken by a growing number of research groups in recent years. Here, we attempt to organize those studies and provide a framework for understanding them (Figure 3). Overall, the proposed methods can be categorized as (i) improving WE simulation, and (ii) improving analysis of WE data. In each of these major categories, multiple directions and strategies have been pursued.

3.1 | Optimizing Progress Coordinate Selection and Binning Scheme

Optimizing the progress coordinates and binning scheme—which controls the allocation of WE trajectories in phase space—is the most important practical aspect of WE. Once a system is selected, the physical timescales cannot be changed, but the efficiency of generating trajectories of value for observables of interest can be significantly under user control via the binning scheme and progress coordinates. As an example, the RMSD to a target structure could be sufficient for determining whether transitions occur or not, but it may not work well as a WE progress coordinate for lack of information on key hydrogen bonds, isomerizations, or conformational re-arrangements that might define mechanistic bottlenecks.

Before describing “modern” approaches, we want to emphasize the importance of expert structural knowledge of the system under consideration, that is, the intuitive approach. Constructing initial, intuitive progress coordinates often will be needed for initial WE runs to generate data and/or provide intuition for features useful in more sophisticated downstream optimization. Intuitive coordinates can also play a key role in troubleshooting results from ‘black box’ methods.

Even with hand-crafted, intuitive coordinates, there is flexibility within WE to determine how bins subdivide chosen coordinates, and bins can be updated during WE simulation without biasing dynamical behavior [18]. One approach that has gained popularity is the minimal adaptive binning (MAB) scheme introduced by Chong and coworkers [36]. MAB enables a parsimonious

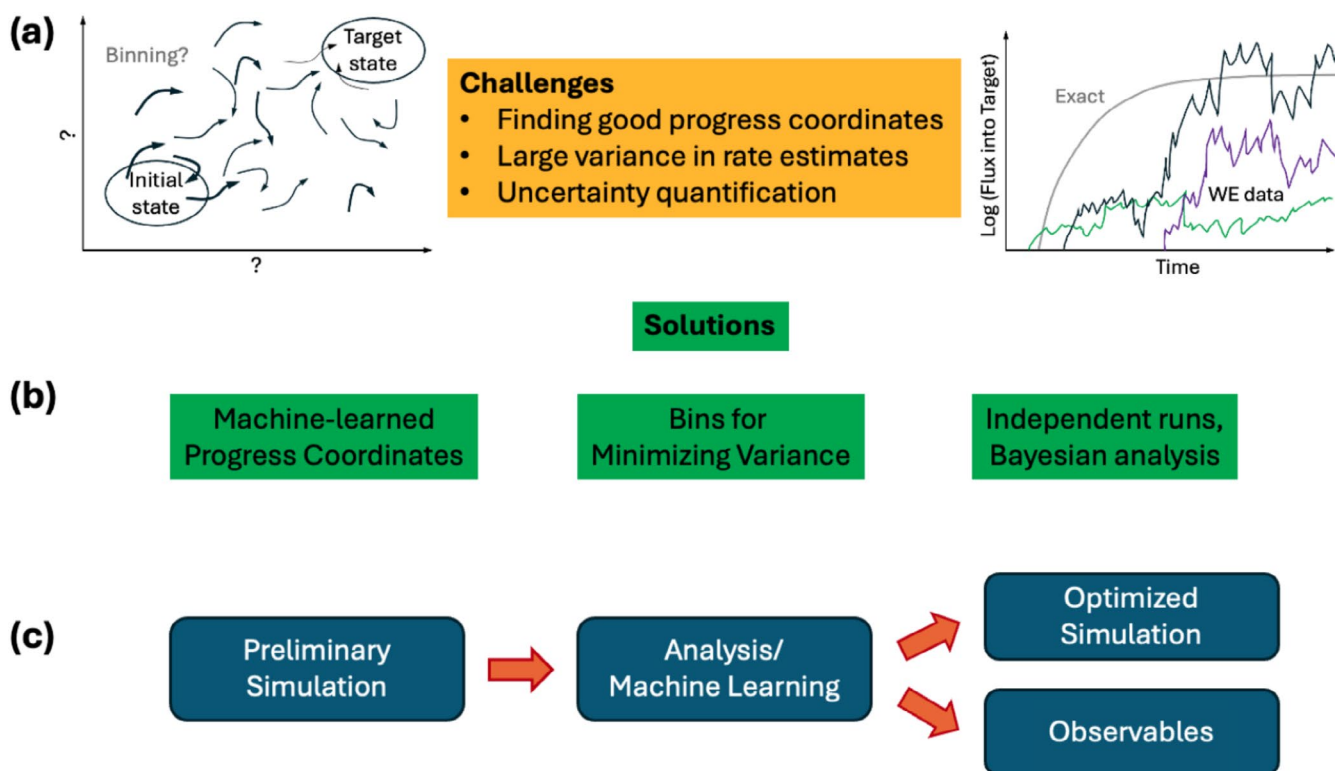


FIGURE 3 | Challenges and solutions in WE methodology. (a) Because of the inherent flexibility of WE, initial user choices for coordinates and other parameters may not be optimal, leading to noisy, large-variance results. (b) Multiple solutions have been proposed for optimizing progress coordinates and bins, as well as for uncertainty analysis. (c) Optimization approaches typically require preliminary simulation data from WE or another method, followed either by additional WE simulation or direct estimation of observables.

use of computing resources by automatically detecting bottleneck regions among WE progress coordinates already sampled, and then assigning additional walkers to those regions and to outliers, while reducing walkers in less important regions. The scheme has been applied in numerous studies [31, 37–45].

A related consideration is whether traditional bins should be used at all. From a mathematical point of view, bins are simply collections of WE walkers employed for resampling [18], and thus demarcated regions of coordinate space need not be defined. Dickson and coworkers have employed groupings of trajectories based on a similarity metric in the “resampling of ensembles by variation optimization” (REVO) scheme that has been applied to study a number of systems [19, 46, 47].

Machine learning (ML), not surprisingly, underpins several approaches proposed for optimizing progress coordinates for WE simulation. For example, Chong, Ramanathan, and coworkers demonstrated that unsupervised learning of progress coordinates can be integrated into a WE simulation by applying a convolutional variational autoencoder [48]. This approach compresses high-dimensional trajectory data down to lower-dimensional representations in latent space, followed by resampling based on outlier trajectories. Tiwary and coworker employed a combination of coordinates, including those from the “state predictive information bottleneck” ML method with those based on expert intuition [49]. The hybrid approach was able to reduce run-to-run variance in WE simulations. Andricioaei and coworkers proposed a linear-discriminant analysis for progress coordinate identification [50].

A variance-reduction principle can be used to define bins with the explicit goal of increasing precision in rate-constant estimates, as proposed by Aristoff and Zuckerman [51]. Mathematical analysis shows that the local first-passage time to the target state is the key coordinate for estimating steady state properties such as the rate constant with minimum variance [52]. This approach has now been applied to atomistic protein folding with impressive results [53]. Relatedly, minimizing the variance of weights within bins can reduce the variance in observables [54], a point which has recently been revisited [55].

3.2 | Optimizing Rate-Constant Estimation

Although in principle rate-constant estimation using Equation (1) should be a straightforward matter of monitoring the flux until steady state is reached [25], in practice two major obstacles are encountered: (i) the relaxation time to reach steady state may be longer than what can be afforded in a WE context when many trajectories are run in parallel and flux recorded in the transient region will generally under-estimate the true flux [56, 57], and (ii) even if the relaxation time is surpassed, flux estimates among WE runs can vary by orders of magnitude (Figure 3), necessitating a large number of replicate runs for a precise rate constant [53, 58]. Solutions have been proposed for both problems.

Two basic strategies have been proposed for estimating a rate constant from transient, pre-steady-state information. One approach is to estimate steady state behavior from the transient

using “history augmented” Markov state models (haMSMs) which can be built from WE data and can provide unbiased estimates for rate constants at any lag time, unlike standard MSMs [28, 57]. Steady-state solution to a haMSM provides the necessary flux value for the rate constant in Equation (1). Relatedly, steady-state estimates from a haMSM can be used to restart a new WE run, check for stability, and iterate this whole process as needed [15]. A different strategy deconvolutes the steady-state behavior from the transient flux using the WE-estimated distribution of t_{TP} values [59]. This strategy corrects for systematic error arising from statistical bias toward shorter t_{TP} values and is most effective when trajectory lengths are comparable to the most probable event duration.

The problem of variance among WE estimates can be addressed via improved binning. For example, an improved choice of progress coordinate that better tracks “kinetic progress” to a target state should reduce variance compared to a more naive choice. Different approaches to improved progress-coordinate selection have been discussed above, along with a global approach for optimizing the binning scheme and walker allocation to minimize variance in rate-constant estimation [52].

3.3 | Quantifying Mechanisms via Path “Fingerprints”

An important biophysical question in molecular simulation studies considers the degree of heterogeneity in the paths of a given process, that is, temporal sequences of conformations in binding or conformational change. As a path sampling method, WE has been a natural tool for probing path heterogeneity [60–62]. However, there is no consensus in the field on how path heterogeneity should be quantified. What is a quantitative “fingerprint” of mechanism in a path ensemble? We therefore alert the reader to recent efforts [63–65] to establish standard tools for analyzing continuous transition trajectories, such as those generated in WE simulation, in the hope that future community-wide efforts will establish standards for this important issue. Related efforts have been made in the MSM community [66, 67], for analysis based solely on a transition matrix that may not have been determined from continuous trajectories.

3.4 | Uncertainty Quantification

Uncertainty quantification (UQ) for WE simulation is challenging for the same reasons that hinder UQ in other types of molecular simulation, namely, undersampling and complex correlation structures [68, 69]. In overview, WE is limited by the intrinsic relaxation times exhibited by the (typically complex) systems of interest, as well as by the nature of the algorithm: the splitting process described above means that numerous trajectories share a partial common history and thus are substantially correlated. Whenever data is correlated, great caution must be used in UQ because many traditional approaches assume statistical independence of values [68, 69].

As a prime example for WE, consider estimating the uncertainty in the rate constant calculated via Equation (1), which explicitly requires simulations to have reached steady state. One

traditional strategy for UQ would employ the time series of flux estimates for auto-correlation or block-averaging analysis [68]; however, the large run-to-run variance in WE simulations suggests time-series analysis could provide flawed UQ because any single WE run is unlikely to be representative. Instead, it has become more standard to perform multiple WE runs and quantify the variation among them [29, 31, 58, 70]. Based on the large observed log-variance among WE runs, that is, rate-constant estimates varying over multiple orders of magnitude, a Bayesian approach may often be appropriate despite limitations in quantifying uncertainty percentiles [71].

3.5 | Estimating Equilibrium Observables

Although WE can fairly be characterized as an intrinsically nonequilibrium method because it tracks trajectories “relaxing” from some initial (typically nonequilibrium) condition similar to “dynamical nonequilibrium MD” [72]. WE can also be used to estimate equilibrium observables such as distributions or conformational free energy differences. A single WE simulation can be used to compute both equilibrium and nonequilibrium observables with a suitable setup and extensive sampling [73]. In principle, WE can be used in several ways to compute equilibrium quantities: (i) perform “equilibrium” WE simulation without recycling boundary conditions for long enough so that it relaxes to equilibrium and directly use the resulting ensemble; (ii) run two nonequilibrium WE simulations with recycling boundary conditions from $A \rightarrow B$ and $B \rightarrow A$, compute rate constants in each direction, and use their ratio to calculate equilibrium population ratios via $p_A/p_B = k_{BA}/k_{AB}$; and (iii) use nonequilibrium data from one or more WE runs to compute a haMSM and solve that for equilibrium quantities [25].

Practical equilibrium calculations with WE require care. Using the relaxation approach (i), it is necessary to start the simulations fairly close to equilibrium (based on preliminary simulations or estimates) and assess the relaxation progress [42, 74]. If a WE run is started very far from equilibrium in a complex system, it is unlikely to have time to relax. An initial estimate for equilibrium can be obtained from MSM analysis of a preliminary WE simulation [25]. When a Markov model analysis of nonequilibrium WE data (iii) is used for post-analysis without additional WE simulation, transition matrix elements obtained from WE data may be biased and require correction by reweighting of trajectory segments in a self-consistent fashion [75]. A simple, uncorrected approach for equilibrium estimation via MSM is available in WESTPA [25, 73].

3.6 | WE Variations and Hybrid Methods

Hybrid methods have been proposed in which WE is used in combination with another approach. Andricioaei and coworkers employed WE to improve the computation of underlying statistics for “milestoning” [38, 76]. Amaro and coworkers employed preliminary simulations using Gaussian-accelerated molecular dynamics to develop initial conditions for WE that benefited from landscape exploration and proximity of the distribution to steady state [77]; they similarly employed MSM techniques for the same purpose [78]. Dinner and workers proposed

approximating the steady state distribution via nonequilibrium umbrella sampling [79] between WE iterations as an alternative to haMSM-based estimates of steady state [15, 57].

4 | WE Software

Advances in WE methodology, combined with highly scalable software implementations, have enabled ambitious simulations of biological processes that are otherwise inaccessible to conventional simulations on typical computing resources. To our knowledge, the only other actively developed WE software packages in the community are wepy [17] and an implementation designed for the Anton special-purpose supercomputer by DE Shaw Research. Here, we highlight advances in WESTPA (The Weighted Ensemble Simulation Toolkit with Parallelization and Analysis), an open-source software package that has been actively developed and maintained for over a decade [9, 15].

4.1 | Scalability

A key strength of WESTPA is its exceptional scalability (i.e., to thousands of CPUs/GPUs) [9, 10], with performance expected to remain efficient until filesystem saturation. The WE strategy is inherently parallel, but optimizing its implementation presents two main challenges. First, as the number of CPUs/GPUs per node increases, managing communication between WESTPA and compute processes becomes more demanding. WESTPA addresses this challenge with a work manager featuring a custom communication layer (over Ethernet or InfiniBand) that maximizes efficiency, scales to hundreds of thousands of workers across thousands of nodes, and minimizes data transmission overhead. Second, the WE algorithm requires a serial checkpoint between iterations for trajectory splitting, merging, and reweighting. Although checkpointing is usually inexpensive compared to dynamic propagation, advanced reweighting schemes—especially those using ML—can significantly extend checkpointing time, leaving most nodes idle. To mitigate this, WESTPA centralizes the storage of raw and processed trajectory data and provides tools for parallelizing computationally intensive reweighting tasks, leveraging both the Python ecosystem and optimized numerical and ML frameworks.

4.2 | Interoperability

WESTPA is designed for modularity, allowing straightforward interfacing with almost any dynamics engine (e.g., MD engines such as AMBER [80], OpenMM [81], and Gromacs [82]) and simulation analysis tools (e.g., MDAnalysis [83], MDTraj [84], LOOS [85], and WEDAP [86]) without requiring modifications to the core software. Since WE is agnostic to physical scale and rigorous for use with any stochastic dynamics method (e.g., MD and Monte Carlo), WESTPA has been successfully used beyond MD, including cell-scale dynamics engines (e.g., MCell [87]) and systems biology packages (e.g., BioNetGen [88]).

Alternatively, WE software can be tightly integrated with a single dynamics engine, allowing the simulation to run continuously with a single initialization call rather than requiring

separate calls at each resampling time interval. This approach is possible for dynamic engines implemented as Python libraries (e.g., OpenMM). For example, wepy [17] is specifically designed to work with OpenMM, reducing the need for frequent file reads and writes. Similarly, in systems biology simulations, where file I/O is particularly intensive, WESTPA has been tightly integrated with BioNetGen [5], streamlining data handling and enhancing performance.

4.3 | WESTPA: Recent and Future Upgrades

WESTPA has undergone major upgrades to handle slower processes (>ms) and larger systems (>1M atoms) [15]. Since WE simulations generate tens of millions of trajectory files, an improved data storage framework has been necessary to facilitate simulation restarts, data sharing, and post-simulation analysis. To address this, WESTPA has expanded on its original HDF5 framework, allowing users to store not only WE data, but also trajectory data in highly compressed, portable HDF5 files (typically fewer than 100 files). These files follow the HDF5 format used by the OpenMM dynamics engine [81] and the MDTraj analysis tool. Additionally, AMBER-based trajectory files, which use the NetCDF format, can be readily adapted to HDF5. Looking ahead, future WESTPA versions will enhance its task distribution framework to (i) reduce latency, (ii) balance computational loads more effectively, and (iii) improve transparency for end users. WESTPA will also integrate with the Dask work manager, an industry-standard solution that would enhance scalability, portability for cloud computing, and fault tolerance—crucial for time-sensitive simulations.

5 | WE Applications—Recent Highlights

In recent years, the WE strategy has advanced multiple fields, including virology, drug discovery, and chemical kinetics. WE has helped fulfill the long-held promise of molecular dynamics simulations by leveraging two key strengths: (i) generating an unbiased path ensemble rich in mechanistic detail, and (ii) directly calculating both overall and step-specific rates in multi-step processes to identify rate-limiting steps. These applications highlight the importance of dynamical effects in understanding the complex biological and chemical mechanisms, spanning from the atomistic level of massive proteins to the quantum-level chemical reactions (Figure 4). Below, we review key accomplishments and insights gained from these applications.

5.1 | Virology: Spike Protein Dynamics

The most ambitious WE application to date has been the atomistic simulation of the SARS-CoV-2 spike protein opening process, focusing on its receptor-binding domain (RBD). This large-scale study (nearly 1M atoms) simulated a seconds-timescale process and was a key component of an international effort that won the 2020 Gordon Bell Special Prize in COVID-19 Research [89]. While rate estimation remains challenging, WE simulation of pathways for the spike-opening process revealed valuable mechanistic insights unattainable through laboratory experiments [10]. Notably, glycans were found to play an active role beyond

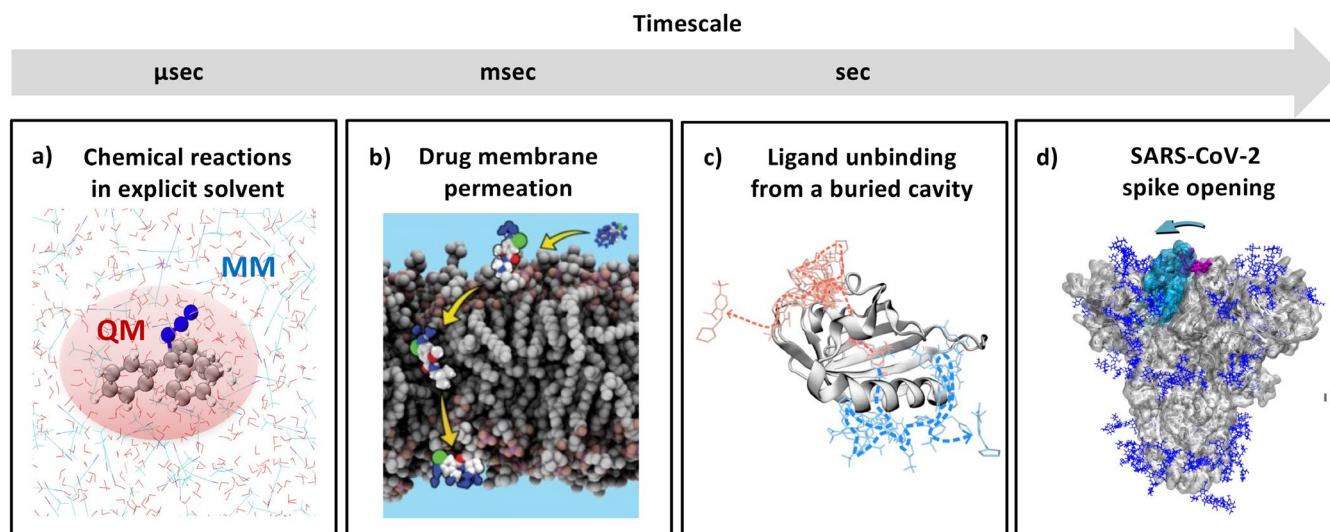


FIGURE 4 | Recent WE applications from the microsecond to second timescale. (a) Microsecond timescale: Hybrid quantum mechanics/molecular mechanics (QM/MM) simulations of chemical reactions [70]. (b) Millisecond timescale: Atomistic simulations of passive membrane permeation by drug-like compounds [44]. (c) Second timescale: Atomistic simulations of a drug-like ligand escaping from an internal receptor cavity, revealing two distinct routes [31]. (d) Second timescale for a million-atom system: Atomistic simulations of SARS-CoV-2 spike protein opening [10].

shielding the virus from the human immune system—specifically, a glycan at N288 acts as a conformational gate for spike opening. This finding was validated by bilayer interferometry experiments and cryo-electron microscopy, which employed the ManifoldEM method [90] to generate large-scale motions of the spike protein that were found to be consistent with those of the simulations. These efforts led to major upgrades to the WESTPA software package, facilitating more efficient WE simulations for slower processes and larger systems [15]. These improvements have since been utilized to compare RBD spike-opening processes across the original, Delta, and Omicron SARS-CoV-2 variants [91] and to support WE-based vaccine design targeting the S2 domain [37].

5.2 | Drug Discovery: Ligand Unbinding and Cryptic Pocket Exploration

One of WE's notable applications in drug discovery is the simulation of protein–ligand unbinding processes on the seconds timescale and beyond—thus far, limited to uncharged ligands [92–94]. The mechanism of such processes has been of great interest to the pharmaceutical industry, given that drug activity often correlates with the inverse of the off-rate, that is, the residence time of the drug in the receptor binding site [95]. Among such processes is the age-old question of how a drug escapes the buried cavities of protein drug targets that are otherwise undruggable. WE simulations successfully captured unbinding pathways of a drug-like, uncharged ligand from the HIF2 α PAS domain, a key cancer target [31]. This domain contains an internal cavity large enough to accommodate various small molecules, including the therapeutically used inhibitor belzutifan [96–99]. The simulations not only yielded an off-rate estimate, but also pathways that revealed two distinct ligand exit routes. These pathways were generated in a blind manner without requiring any previous knowledge of the unbinding process. In contrast, faster, more approximate methods involving either the application of external biasing forces (steered MD) [100] and/

or modification of the energy landscape (metadynamics) [101] require predefined pathways, for example, for water exiting/re-entering the buried cavity [102]. The WE-generated unbinding pathways differ from those generated by these methods, identifying previously unknown conformational gating residues, validated via NMR kinetics [31]. In another drug discovery application, WE has sampled cryptic pockets that are not visible in experimental structures of a protein target, providing potential viable routes to drug design for otherwise undruggable targets [41, 43].

5.3 | Drug Membrane Permeability: A Virtual Assay for Bioavailability

Another significant WE application is the development of a “virtual assay” for predicting drug permeability across lipid membranes. Passive permeability is a crucial property for assessing drug absorption, distribution, metabolism, excretion, and toxicity (ADME/Tox). As a proof of principle, WE simulations assessed the permeability for a set of drug-like amines of varying size, shape, and flexibility through a model 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC) lipid bilayer [44]. The results yielded permeability coefficients consistent with experimental values from the MDCK-LE cell line and Parallel Artificial Membrane Permeability Assays (PAMPA) while also offering mechanistic insights into the transport process. Importantly, despite using a progress coordinate (z -position in the membrane), deemed suboptimal by other methods [103, 104], WE successfully generated pathways and rate estimates with orders of magnitude less computing cost than conventional MD. Thus, in this instance, the WE strategy is much less sensitive to the choice of a poor progress coordinate than free energy-based methods. The WE workflow is available on OpenEye Scientific's Orion cloud computing platform for use on Amazon Web Services. WE has also been applied to peptide membrane permeation [105] and lipid phase separation thermodynamics [42].

5.4 | Chemical Reactions: Mechanistic Insights From Hybrid QM/MM Simulations

WE has also enabled the simulation of chemical reactions in solution, opening up new opportunities to explore how dynamics influence reaction mechanisms. Although other path sampling methods, such as transition interface path sampling [11], have been applied in this context [106–108], their relative advantages have yet to be established. A recent proof-of-principle study demonstrated the power of WE by examining a series of textbook azide-clock reactions, in which an azide anion reacts with trityl cations in acetonitrile-water solution [70]. Using hybrid QM/MM models, WE simulations generated reaction pathways and estimated rates within error of experiment. Notably, these simulations challenged previous assumptions [109]: none of the reactions were diffusion-controlled. Instead, each reaction was limited by the activation step—the rearrangement of an ion-pair intermediate into the product. The study further revealed that increased azide “crawling” among the cation’s phenyl rings correlated with slower reaction rates. Together, these results highlight the unique strengths of WE for path sampling with hybrid QM/MM models and for providing mechanistic insights through kinetics analysis.

6 | Future Directions and Outlook

The WE strategy continues to push boundaries across multiple disciplines by enabling efficient sampling of fully continuous pathways for rare-event processes and facilitating direct rate calculations. The extensive sampling provided by WE offers valuable opportunities to directly assess the accuracy of current force fields in predicting dynamical observables—an aspect that has historically received limited attention in force field development [110–112]. Recent advances have expanded the applicability of WE to areas such as virology, drug discovery, permeability assessment, and reaction dynamics, establishing it as a powerful tool for investigating complex biomolecular and chemical systems.

Despite these advancements, WE has not reached its full potential. The outlook for future WE simulations is highly promising. At the quantum level, WE simulations have the potential to significantly advance hybrid QM/MM methodologies, enabling the exploration of reaction pathways and rates on multi- μ s timescales or longer. This capability is particularly relevant for studying chemical reactions occurring in solution or catalyzed by enzymes, where the QM region involves fewer than 50 atoms treated with semi-empirical methods [70]. Furthermore, the efficiency of such simulations might be further enhanced with the use of adaptive string methods to identify minimum free energy paths as progress coordinates [113, 114]. At the atomistic level, WE simulations can generate pathways and achieve order-of-magnitude rate estimates for therapeutically relevant processes beyond the seconds timescale. Furthermore, the path ensembles generated by WE and similar strategies provide detailed mechanistic insights, including the relative fluxes of competing pathways. These insights could be crucial in understanding how alterations—such as cancer-associated mutations in the DNA-binding domain of tumor suppressor p53—influence flux through specific routes, as investigated in interrupted folding kinetics experiments [115]. Looking ahead, WE strategies are

poised to simulate functionally relevant processes in increasingly large and complex systems, particularly as more structural data becomes available through advanced electron microscopy techniques. These developments will enable studies of environments such as respiratory aerosols [45] and bacterial [116, 117] or human cells [118], further expanding the reach and impact of WE simulations.

Author Contributions

Lillian T. Chong: conceptualization (equal), funding acquisition (equal), writing – original draft (equal), writing – review and editing (equal). **Daniel M. Zuckerman:** conceptualization (equal), funding acquisition (equal), writing – original draft (equal), writing – review and editing (equal).

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Conflicts of Interest

L.T.C. serves on the scientific advisory board of OpenEye Scientific Software (Cadence). The other author declares no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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